

SEMINAR

Defects classification with Modified PatchCore, DINOv3 and VLM

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Motivation: Overcoming VLM Limitations

Previous Approach:

- Feeding defect images directly into VLM
- **Issue:** Classification "hallucinates" when working with precise spatial, geometric, and quantitative reasoning without prior context.

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Proposed Solution:

- A **Neuro-Symbolic** architecture.
- Provide quantitative structural context (size, shape) to VLM *before* requesting a classification.
- Constrain VLM's output to a strict set of ground-truth labels to enable fully automated evaluation.

New Pipeline

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